11. Requirements Manual for “buy” (CotS) Parts

For our sunny-side-up egg frying machine project, a Make-or-Buy analysis was conducted to determine which components should be manufactured in-house and which components should be bought externally as Commercial off-the-Shelf parts. In this strategic decision-making process we considered factors such as reliability in operation, design complexity, hygienic design, regulatory requirements and easy maintenance in a commercial kitchen environment.

Based on this analysis, several components were identified as suitable for external procurement. These parts are already available on the market with defined technical specification, proven performance and standardized interfaces. Buying them reduces development time, lowers manufacturing complexity and supports faster assembly of the machine.

The main criteria for selecting Buy parts are:

- Standardization: Using standardized components reduces design complexity and makes it easier to replace and maintain these parts.
- Proven reliability: Commercially available components are already tested and thus reduce the risk of failure during operation.
- Cost-Effectiveness: Mass-manufactured parts benefit from economies of scale.
- Regulatory Compliance: Pre-certified parts comply with legal safety standards.
- Technical suitability: The selected parts have to meet the technical requirements necessary to ensure reliable operation of the SunnyPro egg-frying machine.

This manual presents the Buy Requirements Table for the selected purchased components. Its purpose is to support the purchasing department in making a possible decision on purchasing a part based on the complete technical requirements and short explanations of their origin and reason.

11.1. List of Components

11.1.1. Power Supply System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin**
1	IEC C20 Power Inlet	1. Connector type: IEC C20 male inlet 2. Current rating: ≥16A 3. Voltage rating: ≥250V AC	Required as standardized detachable mains connection for a commercial 230 V machine.

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2	16A MCB- TYPE B Circuit Breaker	1. Current rating: 16A 2. Voltage rating: 230/400V AC 3. Pole configuration: 2P 4. Suitable standard certification	Protects the mains input wiring and allows safe disconnection of live and neutral conductors.
3	RCCB / RCD-TYPE A (30mA)	1. Residual current sensitivity: 30mA 2. Rated current: ≥16A 3. Voltage rating: 230V AC 4. Pole configuration: 2P 5. Standard compliance: IEC/EN 61008-1	Provides personal protection against electric shock on the AC mains side.
4	Emergency Stop Switch	1. Normally closed contact 2. Red mushroom-head actuator 3. Latching function with manual reset 4. Contact rating suitable for 24V DC control circuit	Needed so kitchen staff can immediately stop the machine in an unsafe situation.
5	DIN Terminal Blocks (L/N/PE)	1. Separate terminals for L, N, and PE 2. Rated current ≥16A 3. Rated voltage ≥250V AC 4. Compatible with 2.5 mm² wire , The selected AC wiring size is 2.5 mm².	Organizes and safely distributes mains wiring inside the electrical cabinet.
6	Protective Earth Terminal Block	1. Compatible with ≥2.5 mm² conductor 2. Bright color identification 3. Low-resistance connection 4. Secure screw or spring connection	Required to connect exposed conductive parts to protective earth for fault protection.
7	AC Wire	1. Conductor cross-section: 2.5 mm² 2. Voltage rating suitable for 230V AC 3. Color coding: brown, blue, green/yellow 4. Heat-resistant insulation	Carries the 16 A mains supply safely and allows clear identification of L, N and PE.
8	24V Industrial SMPS 60W (2.5A at 24V)	1.Input voltage: 230V AC 2.Output voltage: 24V DC, 2.5A 3.Overload and short-circuit protection	Provides the main industrial control voltage for the machine.
9	24V → 12V LM2596T-12	1. Input voltage: 24V DC 2. 3 A continuous (LM2596 class) 3. High efficiency 4. Overload and short-circuit protection	Supplies 12 V loads such as fans or actuators from the 24 V control supply.
10	12V → 5V LM2596T-5	1. Input voltage: 12V DC 2. Output voltage: 5V DC, 2A continuous 3.Output ripple/noise: ≤50 mV peak-to-peak preferred, ≤100 mV peak-to-peak acceptable	Supplies 5 V sensors, logic modules and interface electronics reliably.

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11	24V → 6V LM2596S-ADJ (Adjusted to 6V)	1. Input voltage: 24V DC 2. Output voltage: 6V DC, 2A continuous 3. Low heat generation / high efficiency	Supplies the 6 V servo/actuator circuit without overloading the main 24 V supply.
12	5V → 3.3V Regulator AMS1117-3.3	1. Input voltage: 5V DC 2. Output voltage: 3.3V DC 3. Output current : 800mA 4. Output ripple/noise: ≤50–100 mV peak-to-peak 5. Operating temperature: at least 0°C to 70°C	Supplies 3.3 V controller and logic components.
13	DC Terminal Blocks	1. Separate terminal groups for 24V, 12V, 6V, 5V, 3.3V, and 0V/GND 2. Current rating for 24V/12V actuator branches: ≥10A per terminal 3.Current rating for 5V/3.3V logic branches: ≥2A per terminal 4.Voltage rating: ≥24V DC 5. Wire compatibility: at least 0.5–2.5 mm²	Prevents mixing of different DC voltage levels and simplifies maintenance.
14	Inline Blade Fuse Holders	1. Fuse type: standard automotive blade fuse, ATO/ATC type 2. Voltage rating: ≥32V DC 3. Current rating: 3–5 A for actuator/fan branches; 1- 3 A for logic/sensor branches sized 4. Wire size compatibility: approx. 0.75–2.5 mm²	Allows simple branch protection for DC actuator, fan and logic circuits.
15	Blade Fuse Assortment	1. Fuse type: ATO/ATC blade fuses 2. Voltage rating: ≥32V DC 3. Fuse ratings: 1A, 3A, 5A, 10A	Provides replaceable protection for different DC load branches.
16	Shielded Signal Cable	1.Cable type: shielded multi-core signal cable 2.Number of cores: minimum 3 cores 3. Conductor size: approx. 0.14–0.5 mm 4. Voltage rating: ≥30V DC	Reduces interference on sensor and communication signals near motors, solenoids and converters.
17	Ferrules and Crimp Terminals	1. Ferrule size range: 0.14–2.5 mm² 2. Insulated ferrules 3. Crimp terminals compatible with screw/spring terminal blocks 4. Material: tinned copper conductor sleeves	Ensures secure and serviceable wire connections in the electrical cabinet.

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18	PCB Screw Terminal Connectors	1.Connector type: PCB mount screw terminal block, wire-to-board 2. Pitch: 5.08 mm 3. Voltage rating: ≥30V DC 4. Current rating: ≥5A per contact 5. Wire size compatibility: 0.14–1.5 mm²	Connects external wires to the control PCB safely and replaceably.
19	Electrical Cabinet Cooling Fan	1. Operating voltage: 12V DC 2. Quantity: 2 fans 3. Fan size: 80 x 80 x 25 mm 4. Airflow: approx. 58 m³/h per fan 5. Operating temperature: ≥ -10°C to +70°C 6. Protection: fan guard required	Connects external wires to the control PCB safely and replaceably.
20	Main Microcontroller STM32H743ZITx	1.Logic voltage: 3.3V 2.GPIO available: 114 (LQFP144) sufficient 3.PWM channels needed : ≥ 4 4.UART : ≥ 2 5.SPI : ≥ 2 6.I²C : ≥ 1 7.Operating Temp: 0–70 °C commercial; consider industrial -40 to +85 °C if cabinet ambient gets to ~50 °C	Controls sensors, actuators, HMI communication and machine timing.
Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	DIN Rail	1. Rail type: TS35 / 35 mm DIN rail 2. Material: zinc-plated steel or aluminum	Mounting base for protection devices, terminal blocks and power supply components.
2	Cable Glands / Strain Reliefs	1. Thread sizes: M12, M16, and M20 selected 2. Protection rating: IP54 minimum, IP65 preferred 3. Strain relief function included	Protects cable entries against pulling forces, dust, moisture and grease.
3	Cable Duct / Wire Raceway	1. Type: slotted wiring duct with removable cover 2.Material: flame-retardant PV 3. Separate routing possible for power and signal cables	Keeps cabinet wiring organized and reduces interference and service errors.
4	Cable Ties + Mounts	1.Cable tie material: nylon PA66 2. Tie size matched to cable bundle diameter 3. Temperature rating: at least -10 - +70°C	Fixes cable bundles safely inside the electrical cabinet.

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5	Pan Head Cross Recess Screw	1. Component type: pan head cross recess screw 2. Standard: ISO 7045 3. Screw size: M4 × 8 mm 4. Drive type: cross recess Z	Used for fastening components in the electrical cabinet.

11.1.2. Storage Unit and Breaking Mechanism System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Gate Actuators	1. Actuator type: Tubular pull solenoid, JF-0530B 2. Rated voltage: 12V DC 3. Stroke length: 10-20 mm preferred 4.Actuation Force: ≥10 N pull force 5. Duty cycle: Rated intermittent duty ≤25%	Used to release one egg at a time and prevent double-feeding in the storage mechanism.
2	Flow Sensors	1. Sensor type: IR break-beam sensor pair (emitter + receiver), Adafruit 2167 2. Supply voltage: 3.3-5V DC 3. Detection function: detects individual egg movement 4. Detection range: Detection range up to ~100 mm 5. Response time: fast	Required to confirm egg movement and control the gate sequence reliably.
3	Flip Actuator	1. Actuator type: high-torque metal-gear hobby servo, MG996R 2. Operating voltage: 4.8V-7.2V DC 3. Torque rating: approx. ≥1.0 N·m at 6 V 4. Control method: PWM servo control 5. Rotation range: ≥90° usable travel	Provides controlled angular movement for the flipping/cracking mechanism.
4	Platform Solenoid	1. Actuator type: Tubular pull solenoid, JF-0837B (24 V variant) 2. Stroke length: ~10 mm 3. Force output: ≥25 N pull force 4. Duty cycle: Rated intermittent duty ≤25% 5. Return action: spring return or compatible external return spring	Drives the spear vertically downward to puncture the eggshell, releasing contents onto the cooking plate.

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5	Wedge Solenoid	1. Actuator type: Tubular pull solenoid, JF-0530B (12 V variant) 2. Stroke length: ~10 mm 3. Force output: ≥10 N pullforce 4. Response time: Mechanical stroke time ~25 ms 5. Duty cycle: intermittent operation	Opens the cracked eggshell so the egg contents can drain onto the cooking plate.
6	Solenoid Driver	1. Driver type: multi-channel MOSFET driver module or equivalent switching circuit 2. Load voltage: 12V 3. Logic input: 3.3V or 5V compatible 4. Current rating: minimum 2A continuous per channel, 5A peak per channel preferred	Required because the microcontroller cannot directly switch the higher current needed by solenoids.
7	Wire Harness	1. Harness type: sleeved mixed-gauge wiring assembly 2. Power conductors: 0.75–1.5 mm² 3. Signal conductors: 0.14–0.5 mm² 4. Insulation: flexible insulated wire, rated at least 60V DC and 70°C	A sleeved harness keeps the subsystem wiring grouped, organized, protected and easier to install.
Mechanical Components			
No	**Component**	**Technical Requirements**	**Origin + Reason**
1	Axle Shaft	1. Material: Aluminum 6061 rod 2. Diameter: Ø10 mm 3. Length: 100 mm 4. Straightness: maximum runout/straightness error ≤0.2 mm over the shaft length preferred	Supports rotating or guided mechanism parts with low weight and sufficient stiffness.
2	Spear	1. Material: SS304 stainless steel spear assembly 2. Shaft diameter: Ø3 mm stock shaft 3. Length: 15 mm 4. Tip geometry: conical or tapered tip with nominal tip width of 0.2 mm	Punctures the eggshell and must be corrosion-resistant, cleanable and food-contact suitable.
3	Return Spring	1. Spring type: stainless steel compression spring 2. Spring rate: 0.5–2.0 N/mm 3. Working stroke: 10–30 mm, 4. Fatigue life: minimum 100,000 cycles	Returns the moving mechanism to its starting position after actuation.

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4	Thermal Insulation	1. Material: XPS foam insulation layer 2. Thermal conductivity: ≤0.035 W/m·K preferred 3. Thickness: approx. 20 mm 4. Temperature suitability: at least till 70°C 5. Moisture resistance: closed-cell structure with water absorption	Reduces heat transfer from the cooking area to the storage and breaking mechanism.
5	Inner Lining	1. Material: food-safe aluminum sheet lining 2. Thickness: 0.8–1.5 mm 3. Fabrication: bendable sheet with bend radius of at least 1–2× sheet thickness 4. Corrosion resistance: suitable for humidity	Creates a cleanable inner surface and protects insulation and housing parts.
6	Ventilation Grille	1. Material: ABS ventilation grille 2. Size: 80 mm standard fan grille 3. Open area ratio: minimum 60% open area 4. Slot or mesh spacing: maximum opening width ≤8 mm 5. Thickness: approx. 2–4 mm	Allows airflow while protecting the fan area from accidental contact and larger debris.
7	Fasteners Kit	1. Material: stainless steel fasteners, preferably A2 / AISI 304 2. Screw sizes: mainly M3-M6 3. Thread standard: ISO metric coarse threads 4. Serviceability: removable screw connections preferred	Used for mounting sensors, actuators, panels and mechanism parts while allowing service and cleaning.

11.1.3. Oil System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Peristaltic Pump	1. Pump type: electrically driven peristaltic pump suitable for food-grade oil transfer 2. Rated voltage: 12V DC 3. Flow rate: ~30–60 mL/min 4. Tubing material: compatible with 4 mm ID × 6 mm OD food-grade silicone tubing 5. Duty cycle: intermittent operation	To dispense small, controlled amounts of oil onto the conveyer belt.

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2	MOSFET Driver Module	1. Driver type: MOSFET switching module 2. Load voltage: 12V DC 3. Logic input voltage: 3.3V or 5V 4. Current rating: minimum 2A continuous 5. Control capability: supports ON/OFF switching	A MOSFET driver is needed because the microcontroller cannot directly supply the current required by the pump motor.
Mechanical components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Oil Tank	1. Material: food-grade stainless steel, preferably AISI 304 or 316 2. Tank capacity: 600mL 3. Opening: removable lid preferred 4. Outlet connection: bottom outlet with food-safe fitting; vented cap on top	Stores cooking oil safely and allows refilling, cleaning and reliable oil supply to the pump.
2	Inline Check Valve	1. Valve type: inline one-way check valve for oil line 2. Material: food-safe and oil-resistant body 3.Tubing connection: 4 mm push-fit or barbed 4. Flow direction marking: clear arrow or marking on valve body	1-Prevents oil backflow into the tank when the pump stops, and prevents drip from the nozzle between doses (which would otherwise cause the next egg to land in a pool of stale oil).
3	Silicone Tubing	1. Material: food-grade silicone tubing 2.Inner diameter 4 mm, outer diameter 6 mm 3. Oil compatibility: resistant to cooking oil 4. Flexibility: bend radius approx. 3x	Transports cooking oil safely from tank to pump and nozzle without contamination or leakage.
4	Flat Fan Spray Nozzle	1. Nozzle type: flat fan oil spray nozzle 2. Spray angle: 65° 3. Flow rate: ~48 mL/min 4. Material: stainless steel or food-grade plastic 5. Orifice size: approx. 0.2–0.5 mm	Spreads the oil evenly over the frying surface instead of dispensing it as a single stream.
5	Hose Adapter	1. Fitting type: hose adapter 2. Material: stainless steel or food-grade plastic 3. Tubing compatibility: 4 mm ID silicone tubing 4. Nozzle connection: "1/8" NPT 5. Leak sealing: compatible with O-ring	Connects the flexible oil tubing to the spray nozzle without leakage.

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6	Nozzle Mounting Bracket	1. Bracket type: adjustable mounting bracket 2. Material: stainless steel or aluminum 3. Angle adjustment: allows spray angle adjustment of approximately ±15° to ±30° 4. Mounting holes: fixing holes compatible with M3–M5 screws	Holds the nozzle in the correct position for even oil distribution over the frying area.
7	Pump Vibration Mounts	1. Mount type: rubber vibration isolators or rubber grommets 2. Material: oil-resistant rubber, e.g. nitrile rubber NBR or EPDM 3. Load capacity: rated for at least 1–3 kg 4. Deflection: approximately 1–3 mm compression deflection under pump load preferred	Rubber vibration mounts are used because the pump motor can create vibration during operation that should not transfer directly into the machine frame.

11.1.4 Screen System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Touchscreen Display Module — Nextion NX8048P070_011R	1.Type: 7" HMI touchscreen module 2.Display: 7", 800 × 480 px, 16-bit colour 4.Supply voltage: 5 V DC 5.Supply current: ~430-510 mA 6.Communication: TTL UART (5 V logic) 7.Operating temperature: -20 °C to +70 °C	Single integrated HMI module chosen over a discrete-component display because the Nextion handles GUI rendering, touch decoding, and backlight management internally, the main MCU only needs to send/receive text commands over UART. This eliminates the need for a second MCU, FPC connectors, touch controller, backlight driver, and etc.
2	JST-XH 4-pin Connector	1.Type: 4-pin JST-XH connector, 2.5 mm pitch 2.Pin assignment: 1 = +5 V, 2 = TX (Nextion-side), 3 = RX (Nextion-side), 4 = GND 3.Current rating: ≥ 1 A per contact 4. Keyed/polarised connector preferred	Connects the touchscreen module to the main PCB for power supply and UART communication.
Mechanical Components			

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No.	Component	Technical Requirements	Origin + Reason
1	Mounting Hardware	1.Type: M3 screws + standoffs (brass or stainless) 2.Screw length: 6–10 mm 3.Standoff length: 5–10 mm 4.Material: brass or stainless preferred	Fixes the touchscreen securely to the front panel while keeping clearance behind the display PCB.

11.1.5. Transportation System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Stepper Motor	1. Motor type: NEMA 17 bipolar stepper motor, double-shaft type 2. Holding torque: approx. 0.75–0.85 Nm 3. Step angle: 1.8° 4. Rated current: approximately 1.0–2.0 A per phase 5. Shaft diameter: 5 mm double shaft	A double-shaft stepper motor is used because the front shaft can drive the conveyor while the rear shaft remains available for an encoder, reference disc, hand adjustment, or future feedback function.
2	Stepper Motor Driver	1. Driver type: external bipolar stepper motor driver 2. Supply voltage: 24V DC 3. Output current: adjustable, minimum 2.0 A per phase 4. Control input: STEP/DIR input compatible with 3.3V logic signals	An external stepper driver is needed because the microcontroller cannot directly supply the motor phase current.
3	Control Wiring and Connectors	1. Wiring type: mixed-gauge wiring 2. Motor conductors: 0.5–0.75 mm²motor phase wires 3. Sensor/control conductors: 0.14–0.34 mm² signal wires 4. Voltage rating: ≥30V DC 5. Insulation temperature rating: ≥ 70°C	Connects motor, driver and sensors safely while preventing wiring errors, heat damage and loose connections.

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4	Hall Effect Sensor	1. Sensor type: A1104-class Hall effect switch sensor, 3-pin type 2. Supply voltage: 5V DC 3. Output signal: compatible with 3.3V controller input 4. Detection distance: approximately 2–5 mm from the 8 x 3 mm SmCo magnet 5. Operating range: ≥ 0-100°C	A Hall effect sensor is used because it detects the home/reference position without physical contact or mechanical wear.
5	SmCo Magnet	1. Magnet type: samarium cobalt SmCo permanent magnet 2. Magnet size: 8 x 3 mm 3. Temperature resistance: ≥150°C 4. Corrosion protection: e.g. coated or protected surface	Acts as the magnetic reference marker for the Hall sensor near the heating zone.
6	Incremental Rotary Encoder	1. Encoder type: incremental rotary encoder with A/B quadrature outputs 2. Resolution: approx. 600 P/R 3. Supply voltage: 3.3V DC 4. Shaft/coupling compatibility: matched to drive shaft diameter (5 mm)	Provides feedback about conveyor shaft movement, speed and direction.
Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Roller Chain with Bent Attachments	1. Chain type: roller chain with K1 bent attachments, DIN ISO 606 / DIN 8187 compatible 2. Pitch: 9.525mm 3. Material: special chain steel or equivalent wear-resistant steel 4. Attachment configuration: one-sided 5. Length: approx. 1500 mm	1- A standardized roller chain is used because the conveyor moves the egg carrier plates through the heating zone and provides mounting points for the cooking plates.
2	2-Bolt Flange Bearing Block	1. Component type: 2-bolt flange bearing block 2. Bearing type/size: KFL000 type or equivalent 3. Bore diameter: 10 mm 4. Material: zinc alloy, cast housing or corrosion-resistant housing 5. Rotation: low-friction bearing	Supports the conveyor drive and idler shafts while keeping the sprockets aligned.

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3	Flexible Clamp Motor Coupling	1. Component type: flexible clamp shaft coupling 2. Bore sizes: 10 mm 3. Material: aluminum or steel coupling body preferred 4. Flexibility: able to compensate small shaft misalignment 5. Torque capacity: ≥1 Nm 6. Length: approx. 20 mm	Transfers torque from the motor/gearbox to the conveyor shaft while tolerating small assembly misalignment.
4	Planetary Gearbox for Motor	1. Component type: planetary gearbox for NEMA 17 stepper motor 2. Gear ratio: 1:27 3. Motor compatibility: compatible with NEMA 17 4. Input/Output shaft compatibility: 5 mm / 10mm 5. Rated output torque: ≥2–3 Nm preferred	1- A planetary gearbox is used to reduce motor speed and increase torque for slow, controlled conveyor movement through the heating zone.

11.1.6. Containment Subsystem

Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Non-stick Surface	1. Component type: food-safe non-stick coating 2. Food safety: suitable for direct food contact 3. Temperature resistance: ≥150°C 4. Thickness: coating thickness approx. 20–50 µm, or replaceable liner thickness approx. 0.1–0.5 mm	1- A non-stick surface prevents the egg from sticking in the frying or transfer area and allows easier cleaning after operation.
2	Food-grade Seals / Gap Strips	1. Component type: food-grade sealing strip or flexible gap cover 3. Temperature resistance: ≥150°C 4. Strip thickness: approx. 1–3 mm 5. Strip width: approx. 10–20 mm	1- A sealing strip or gap cover closes small gaps where egg, oil or food residue could collect, improving hygiene and cleanability.

11.1.7. Egg-Shell Storage

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Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Non-disposable Inner Liner	1. Component type: removable non-disposable protective liner 2. Thickness: approx. 1.0 mm 3. Fit: cut to fit 4. Temperature range: at least 0-100°C	A removable liner protects the drawer interior and simplifies cleaning after shell waste collection.
2	Liner Retaining Strips / Clips	1. Component type: retaining clips or strips 2. Material: SS304 stainless steel or food-grade PE 3. Size: for strips approx. 80-100 mm long and approx. 0.8-1.5 mm thick, for chips approx. 15-25 mm (min. Needed 4-6 per machine)	1- Retaining clips or strips keep the silicone liner in position, while still allowing drawer removal during cleaning.
3	Rod End Collars / Retaining Stops	1. Component type: shaft collar or retaining stop for guide rods 2. Rod compatibility: suitable for Ø8 mm guide rods 3. Material: SS304 stainless steel	Retaining stops are needed to prevent the guide rods or pusher assembly from moving beyond the intended travel range.
4	Drawer Latch / Stop	1. Component type: stainless steel drawer latch, spring latch, or mechanical stop 2. Holding force: approx. 10–30 N 3. Size: compact latch body, approx. 20–50 mm length 4. Durability: suitable for repeated opening and closing cycles, ≥10,000 cycles preferred	Keeps the shell-waste drawer securely closed during operation while still allowing manual opening for emptying.
5	Front Sealing Strip / Gap Cove r	1. Component type: food-grade silicone sealing strip 2. Strip thickness/width: approx. 1–3 mm / approx. 10–20 mm 3. Length: cut to fit drawer front or inlet gap; approx. 260 mm	A sealing strip or gap cover is needed to reduce shell dust, residue, smell, and small splashes escaping through the drawer front or inlet gap.
6	Stainless Fasteners Kit	1. Material: SS304 / A2, stainless steel fasteners 2. Screw sizes: mainly M3-M5 3. Thread standard: ISO metric coarse threads 4. Length range: approx.6–16 mm	A fasteners kit is used to assemble drawer rails, brackets, handle and support parts while allowing maintenance and replacement.

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		5. Serviceability: removable connections preferred	

11.1.8. Bottom Heating Subsystem

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Cartridge Heaters	1. Component type: electrical cartridge heater 2. Size: Ø8 × 160 mm 3. Power rating: 200–250 W per heater 4. Supply voltage:230V AC 5. Sheath material: stainless steel sheath (Incoloy 800 preferred) 6. Temperature capability: ≥300°C	A cartridge heater is used because the bottom heating block needs compact, direct heat input from inside the block.
2	Temperature Sensor	1. Sensor type: PT100 2. Measurement range: 0-250°C 3. Accuracy: ±1–2°C in cooking range 4. Probe diameter: approx. Ø3–5 mm 5. Probe length: ~30–50 mm	A temperature sensor measures the heating block temperature so the controller can regulate the frying temperature.
3	Temperature Sensor Amplifier	1. Part: MAX31865xAP or equivalent PT100 interface 2. Supply: 3.3 V DC 5. Interface: SPI (SCLK, SDI, SDO, CS) to STM32H743, 3.3 V logic 6. Wiring mode: 3-wire or 4-wire PT100	Converts the PT100 signal into digital temperature data for the main controller.
4	Solid-state Relay	1. Component type: Zero-cross AC SSR; DC-controlled and AC-switched 2. Load voltage: 230V AC output switching 3. Load current rating: preferably 25A 4. Input control voltage: compatible with 3.3V or 5V control signal 5.Heatsink: mandatory.	Allows the controller to switch the heater power safely without directly handling mains voltage.
5	Heater Branch Fuse	1. Type: T10A 250 V slow-blow cartridge fuse, 5 × 20 mm 2. Breaking capacity: ≥1500 A HBC (high breaking capacity)	Protects the heater branch wiring against overcurrent or short-circuit faults.

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6	High-temperature Wiring	1. Component type: heat-resistant heater wiring 2. Conductor cross-section: ≥1.5 mm² 3. Voltage rating: ≥ 300V AC 4. Temperature rating: ≥180°C 5. Insulation material: PTFE or fiberglass-sleeved insulation	High-temperature wiring connects the heaters safely in the hot area where normal PVC wiring could degrade.
7	Grounding Hardware	1. Component type: protective earth bonding hardware 2. Grounding connection: ring terminal or grounding lug 3. Conductor compatibility: 4 mm² PE . Electrical resistance: low-resistance bond, preferably ≤0.1 Ω between metal part and PE connection	Bonds the heating block and metal frame to protective earth for electrical safety.
Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Thermal Compound	1. Component type: heat-conductive thermal paste 2. Temperature range: approx. 0-250°C 3. Thermal conductivity: ≥1 W/m·K 4. Electrical property: electrically insulating preferred	Thermal paste is needed to improve heat transfer where small air gaps may exist between the sensor and aluminum heating block.

11.1.9. TOP IR Heating System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	IR Heater Element	1. Heater type: shielded IR heater or quartz/ceramic IR element 2. Heated length: approx. 400–450 mm 3. Power rating: 300–600 W 4. Supply voltage: 230V AC	An IR heater element provides top-side radiant heat so the egg can be cooked evenly from above while moving through the heating zone.
2	IR Power Control / Relay	1. Component type: relay or solid-state relay 2. Load voltage: 230V AC 3. Load current rating: ≥ 10A 4. Input control voltage: 5V DC control signal	A relay or SSR is needed so the controller can switch the top IR heater power safely.

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3	Thermal Cutoff / Safety Sensor	1. Component type: thermal cutoff switch, thermal fuse or safety temperature sensor 2. Cutoff / alarm temperature: approx. 180–250°C 3. Signal type: normally closed contact preferred, opens during over-temperature fault 4. Reset type: manual reset	A thermal cutoff or safety sensor protects the top heating area if the IR heater overheats or the control system fails.
4	PT100 Sensor	1. Sensor type: PT100 resistance temperature sensor 2. Measurement range: 0-250°C 3. Accuracy class: Class B minimum 4. Probe diameter: approx. Ø3–5 mm 5. Connection type: 3-wire preferred	A PT100 sensor monitors the temperature near the IR heater module for temperature control and safety supervision.

11.1.10. Interface/Integration System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	SSR Heat Sink	1. Component type: aluminum heat sink for SSR 2. Compatibility: approx. 58 × 45 mm 3. Thermal resistance:≤2.5–3.0 °C/W 4. Operating temperature: ≥70°C	Removes heat from the SSR during heater switching and prevents overheating in the electrical enclosure.
2	Ceramic Terminal Block	1. Component type: ceramic high-temperature terminal block 2. Voltage rating: ≥250V AC 3. Current rating: ≥10A 4. Number of poles: 2-pole 5. Wire compatibility: suitable for approx. 0.75–1.5 mm² heat-resistant wires 6. Temperature resistance: ≥ 200°C	Provides a heat-resistant connection point for 230 V heater wiring near the heating area.
3	Low-voltage Terminal Blocks	1. Component type: low-voltage terminal blocks for sensor and control wiring 2. Voltage rating: ≥30V DC 3. Current rating: ≥2A per terminal 4. Number of poles: 4 pole 5. Wire compatibility: suitable for approx. 0.14–0.75 mm² signal/control wires	Organizes low-voltage sensor, control and driver wiring and reduces wiring mistakes during assembly or maintenance.

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11.1.11. Nuts/Bolt/Washers for Conveyer System, Top/Bottom Heating and Main Housing System

Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Countersunk Raised Head Cross Recess Screw	1. Standard: ISO 7047 2. Screw size: M10 x 25 mm and M10 x 60 mm 3. Drive type: cross recess Z 4. Material: stainless steel or corrosion-resistant steel preferred	A countersunk raised head screw is used where a low-profile screw head is required for strong fastening points in the housing or heating/conveyor structure.
2	Pan Head Cross Recess Screw	1. Standard: ISO 7045 2. Screw size: M8 x 16 mm, M4 x 30 mm and M3 x 20 mm 3. Drive type: cross recess Z 4. Material: stainless steel or corrosion-resistant steel preferred	Used for general fastening points where a flush countersunk head is not required.
3	Socket Head Cap Screw	1. Standard: ISO 4762 2. Screw size: M6 x 20 mm and M3 x 10 mm 3. Material: stainless steel preferred 4. Head type: cylindrical socket head / Allen head	Used for stronger or more compact mechanical fastening points, especially in the conveyor and support structure.
4	Prevailing Torque Hex Nut	1. Standard: ISO 7042 2. Thread size: M6 3. Thread standard: ISO metric coarse thread M6 × 1.0 4. Locking function: self-locking design	Prevents loosening of M6 screw connections caused by conveyor movement or vibration.
5	Plain Washer	1. Standard: ISO 7089 2. Washer size: M6 3. Inner diameter: suitable for M6 screw clearance	A plain washer is used to distribute the fastening load under the screw head or nut.
6	Socket Set Screw, Cup Point	1. Standard: ISO 4029 2. Screw size: M5 × 20 mm 3. Thread standard: ISO metric coarse thread M5 × 0.8	A socket set screw is used because the sprocket or hub must be fixed to the shaft without using a large external fastener.
7	Hex Thin Nut	1. Standard: ISO 4036, Grade B 2. Thread size: M3 and M4	A hex thin nut is used where a compact M3 fastening connection is needed.

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		3, Thread standard: ISO metric coarse thread	

11.1.12. Rotating Plate System

Electrical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Stepper Motor	1. Motor type: 5V geared stepper motor, 28BYJ-48 class or equivalent 2. Step angle / gearbox: geared stepper motor with small step movement 3. Output torque: ≥30–40 mN*m 4. Speed range: low-speed rotation, approx. 1–10 rpm 5. Duty rating: intermittent operation	A geared stepper motor is used because the serving plate needs controlled step-by-step rotation, so that each cooked egg can be placed in a defined position.
2	Stepper Driver Module	1. Driver type: ULN2003A class or equivalent 2. Motor supply voltage: 5V DC 3. Logic input voltage: compatible with 3.3V controller signals 4. Output current: ≥500 mA total 5. Channels: 4 output channels	Allows the microcontroller to control the geared stepper motor safely.
3	Microswitch	1. Switch type: lever-type microswitch with NO/NC/COM contacts 2. Contact rating: ≥5V DC signal switching 3. Actuation force: approx. 0.5–2 N 4. Mechanical life: ≥ 100,000 cycles	A lever microswitch is used because it gives a simple physical signal when the rotating plate reaches the home position.
4	Load Cell	1. Sensor type: single-point bar load cell 2. Rated capacity: 5 kg 3. Supply / excitation voltage: 5V DC 4. Output signal: millivolt bridge signal approx. 1.0–2.0 mV/V sensitivity 5. Accuracy: ±1–5 g resolution after calibration preferred	A single-point load cell is used because the system needs to detect whether the serving plate is present by measuring weight.

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5	HX711 Amplifier Module	1. Module type: load-cell amplifier and ADC module 2. Supply voltage: 3.3V DC 3. Input compatibility: must be compatible with load cell 4. Data interface: DT/DOUT and SCK interface 5. Sampling rate: selectable or fixed 10–80 samples/s	The HX711-type amplifier is needed to convert the small load-cell signal into readable data for the controller.
6	Wiring and Connectors	1. Wiring type: jumper cables and connector wiring for motor, microswitch, load cell and driver module 2. Motor wiring: 4-core wiring, approx. 0.14–0.34 mm² 3. Sensor wiring: 2–4 core signal wiring, approx. 0.14–0.25 mm², 4. Voltage rating: ≥30V DC	Wiring and connectors are needed to connect all electrical components of the rotating plate system safely and replaceable.
Mechanical Components			
No.	**Component**	**Technical Requirements**	**Origin + Reason**
1	Lazy Susan Bearing	1. Component type: turntable bearing 2. Outer diameter: Ø120 mm 3. Load capacity: ≥5 kg 4. Material: corrosion-resistant material preferred 5. Rotation: smooth 360° rotation	A Lazy Susan turntable bearing is needed because the serving plate must rotate while being supported from below and allowing smooth rotation.
2	Rubber Drive Wheel	1. Component type: rubber or polyurethane friction wheel 2. Outer diameter: Ø38.1 mm 3. Bore diameter: 5 mm 4. Wheel width: 24.4 mm 5. Material hardness: approx. 50–80 Shore A	A friction drive wheel is needed to transfer motor rotation to the serving plate without a direct central shaft connection.
3	Countersunk Flat Head Cross Recess Screws	1. Standard: ISO 7046-1 2. Screw size: M4 x 16 mm and M4 x 8 mm 3. Thread standard: ISO metric coarse thread M4 x 0.7 4. Material: stainless steel, preferably A2 / SS304	Countersunk screws are required so the screw head does not protrude above the bearing or mounting surface.

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4	Socket Head Cap Screw	1. Standard: ISO 4762 2. Screw size: M4 x 10 mm and M6 x 20 mm 3. Thread standard: ISO metric coarse thread 4. Material: stainless steel	A socket head cap screw is used for strong mechanical fastening points for the motor and load-cell plate connections.
5	Pan Head Cross Recess Screw	1. Standard: ISO 7045 2. Screw size: M2.5 x 16 mm M5 x 12 mm 3. Thread standard: ISO metric coarse thread 4. Material: stainless steel	Suitable because the microswitch and the bracket mounting connection does not require a flush countersunk head.
6	Socket Countersunk Head Screw	1. Standard: ISO 10642 2. Screw size: M4 × 30 mm 3. Thread standard: ISO metric coarse thread M4 × 0.7 4. Material: stainless steel	A socket countersunk screw is used because the top load cell plate needs a flush screw head with reliable tightening.
7	Hex Thin Nut	1. Standard: ISO 4036, Grade B 2. Thread size: M2.5, M4 and M6 3. Thread standard: ISO metric coarse thread 4. Material: stainless steel	A hex thin nut is used where the fastening area has a limited height and a normal nut may interfere with nearby parts.

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